

MID-TERM TEST

Course Name: Modelering och Reglering / Modeling and Control	Course Code: R0004E
Test duration: 1.5 hours (13:00-14:30)	Date: 2025-04-17

Allowed aids:

- non-programmed calculator
- documents provided in the "Supplementary Materials" folder on Canvas, except the books.
- an A4-sheet (one side) with own notes (no solutions of any exercises) that are to be submitted with the solution sheets
- "formelsamling i reglerteknik" document

On duty teacher (complete telephone number): Dina Shona Laila, +46 76 049 2718	On duty teacher (complete telephone number):
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Grade scale: 3 – 48 points, 4 – 64 points, 5 – 80 points	Number of questions and total score: 3 questions, 100 points
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Other information:

General instructions

Check that you have received complete question sheets.

All new answers begin on a separate page.

Write the page number of your answer sheets.

Write clearly and do not use red pen.

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1. An aircraft attitude varies in roll, pitch and yaw, as defined in Figure 1. For the stabilization of the roll motion the process goes as follows. The system measures the actual roll angle with a gyro and compares this measured roll angle with the desired roll angle which is used as the reference. The ailerons respond to the roll angle error, by undergoing an angular deflection. The aircraft responds to this angular deflection, producing a roll angle rate.

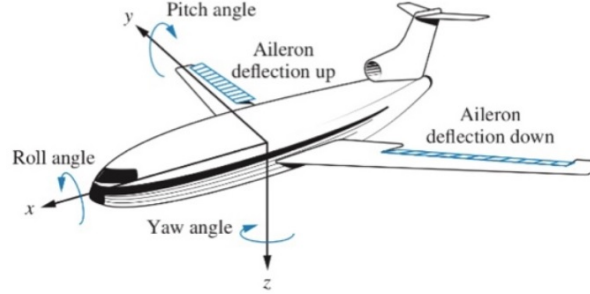


Figure 1: An aircraft attitude defined.

- (a) Draw the closed-loop block diagram of the aircraft system, and identify each element of the block diagram, according to the standard control system block diagram. (15)
- (b) Considering this aircraft control system, determine the state variables, the input variable, the output variable, and the main parameters of the system. (10)

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2. Given the transfer function of a third order plant:

$$G(s) = \frac{1}{(s^3 + 5s^2 + 20s + 16)}$$

- (a) One of the poles of the system is $p_1 = -1$. Determine the other two poles of the system, and find the second order system parameters of the system. (10)
- (b) Find the inverse Laplace transform of the transfer function $G(s)$. (10)
- (c) Determine the steady-state error of the close-loop system $T(s)$, comprising the plant $G(s)$ and a unity feedback, for a unit ramp input. (10)
- (d) Determine the type of the system, and give comments relating to your answer in (c). (5)

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3. Given the dynamic equation of a second order translational mechanical nonlinear system:

$$\ddot{y} + 6\dot{y} + 5\dot{y}^2 + 4y^2 + 4y = u$$

- (a) Defining the output $y(t) = x_1(t)$, show that the linearization of the model at the origin can be written as:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & -6 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

- (b) Using the output definition in (a), write the output equation of the state-space model (determine matrix C and matrix D). (10)
- (c) Find the equivalent transfer function $\frac{Y(s)}{U(s)}$ of the linearized model of the system. (5)
- (d) Find the second order system parameters of the system, and determine the type of response of the system, based on the damping. (10)
- (e) Determine the stability of the system, and give explanation. (5)
- (f) Sketch the unit step response of the system. (5)

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